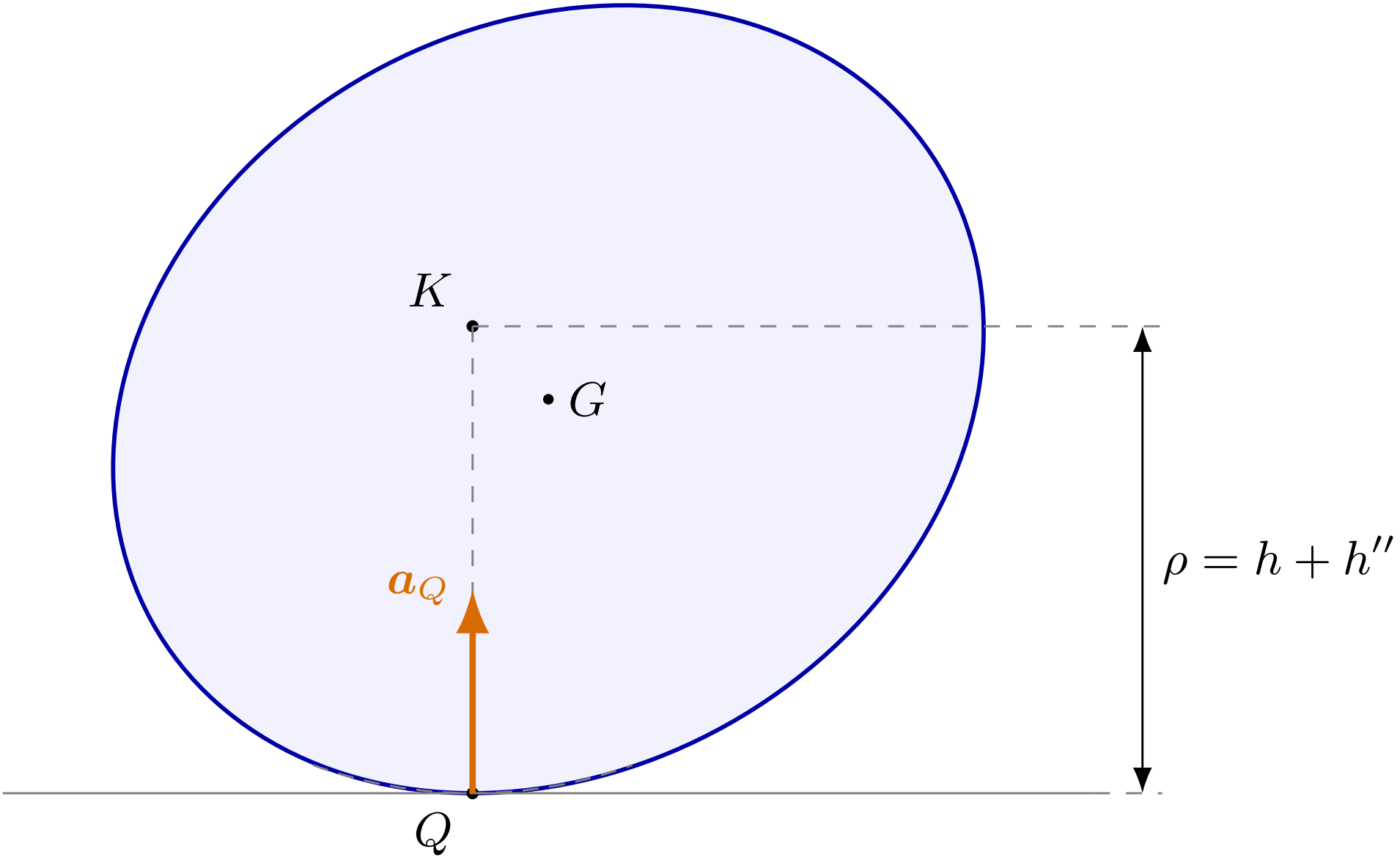

$$\boldsymbol{v}_Q = 0, \quad \boldsymbol{a}_Q = \rho\omega^2 \boldsymbol{e}_y$$